
Sieve: Training-Free, Model-Free, Question-Conditioned Context Compression for Efficient Long-Context LLM Inference

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Abstract

1 We introduce Sieve, a training-free and model-free context compressor that selects
2 a small subset of context sentences before a long-context query is submitted to a
3 large language model. Prefill compute, key-value cache memory, and per-token
4 API fees all grow with context length, yet in question answering over documents
5 most of the context is unrelated to the query; existing compressors either run a
6 neural scorer over the full context, which reintroduces the hosting cost they were
7 meant to remove, or truncate, which discards information unevenly. Sieve scores
8 each sentence with three statistics that are computed in a single pass without any
9 model: an inverse-document-frequency-weighted coverage of query content terms,
10 a U-shaped positional prior that reflects where language models attend reliably,
11 and a redundancy penalty applied greedily under a word budget, with selected
12 sentences emitted in their original order. Measured on QASPER with a produc-
13 tion LLM as the reader, a $4\times$ compression leaves end-task accuracy statistically
14 matched with head truncation (paired 95% interval on the F1 difference [-14.9,
15 8.2]) with point estimates above every question-agnostic selector in F1, and under
16 a reader-free protocol on the benchmark’s annotated evidence Sieve retains more
17 gold evidence than all baselines at every ratio beyond two, with the margin over
18 truncation growing from 8 recall points at $4\times$ to a factor of four at $16\times$ with sepa-
19 rated confidence intervals, while compressing a 6,000-word context in 51.2 ms on
20 one CPU core with zero model parameters, matching the wall-clock of the single
21 scorer pass that perplexity-gated pipelines require while eliminating the model,
22 GPU, and tokenizer they must host, and their coarse-to-fine variants repeat that
23 pass several times. A selector-level analysis attributes the quality to question con-
24 ditioning, exposes an explicit trade-off between positional reliability and middle-
25 of-document recall, and an analytical projection maps the compression onto prefill
26 FLOPs, cache memory, and dollar cost for LLaMA-class deployments.

27 1 Introduction

28 Long context is now a headline capability of large language models, and it is billed by the token.
29 Serving a 32k-token prompt costs the customer proportionally more, costs the provider a prefill pass
30 whose FLOPs scale linearly in prompt length, and occupies key-value cache memory that scales with
31 both length and model depth. In document question answering, retrieval-augmented generation,
32 and agent pipelines, the context that surrounds a query is often one or two orders of magnitude
33 longer than the few sentences that actually determine the answer. Compressing the context before
34 submission attacks all three costs at once.

The default compressor in practice is truncation: keep the first B tokens and drop the rest. Truncation is free, but it interacts badly with the positional biases of long-context models. Evidence placed in the middle of a prompt is attended to least reliably, a phenomenon reported as the lost-in-the-middle effect [9], so truncation that keeps only the head both discards late evidence and concentrates risk on the positions that models read best. The alternative line of work scores tokens with a neural model and keeps the lowest-perplexity or highest-importance ones: LLMingua and its successors gate tokens with a small language model [5, 6, 11], and Selective Context filters by self-information [8]. These compressors recover most task quality at high ratios, but they reintroduce the infrastructure the compression was meant to avoid: a hosted scorer, GPU capacity for at least one forward pass over the full context, a tokenizer matched to the scorer, and in the coarse-to-fine variants several such passes.

This paper asks how far a compressor can go with no neural network at all. The question matters for settings where the scorer is the bottleneck: edge and offline deployment, serverless functions where a 117M-parameter checkpoint is not welcome, private data pipelines that cannot ship text to a scoring endpoint, and high-volume services where the compression itself must cost less than the tokens it saves.

Our answer is Sieve, a sentence-level selector built from three statistics that are cheap, corpus-local, and question-directed. A sentence scores higher when it covers query terms with high inverse document frequency computed over the context itself, when it sits near the beginning or end of the context where models attend most reliably, and when it is dissimilar to sentences already selected; the last term is a redundancy penalty applied greedily under a word budget, which makes the whole rule an instance of maximal marginal relevance [1] specialized with IDF weighting [13] and a positional prior. Selection preserves original sentence order, so the compressed context remains coherent discourse rather than a bag of fragments. The entire pipeline is a few hundred lines of Python with no dependencies beyond the standard library and NumPy.

We evaluate Sieve end to end on QASPER, question answering over scientific papers [3], with a production LLM as the reader. At a $4\times$ compression, end-task F1 of Sieve is statistically matched with head truncation at matched budgets (paired 95% interval [-14.9, 8.2] on the F1 difference) with point estimates above the question-agnostic selectors in F1; under a reader-free protocol on benchmark’s annotated evidence, Sieve retains more gold evidence than every baseline, and the margin over truncation grows as the budget tightens, reaching a factor of four at $16\times$. A placement analysis characterizes how the U-shaped prior allocates retention across document positions, with the reader-side positional profile taken from prior work rather than re-measured under our reader, and a cost study bounds the compression itself at 51.2 ms for a 6,000-word context on one CPU core, against the served floor of the single forward pass any perplexity-gated compressor must pay, with an analytical projection of the savings onto prefill FLOPs, key-value memory, and API cost for LLaMA-2 class deployments using published architecture constants [14]. The end-task evidence covers the $4\times$ operating point; the ratio sweep and the ablations are selector-level, and their scope is stated where they appear.

Our contributions are:

- a training-free, model-free, question-conditioned context compressor whose cost is milliseconds on a CPU core and whose code has no dependencies beyond NumPy;
- an evaluation that couples end-to-end QA accuracy under a production LLM with a reader-free evidence-recall protocol, giving compression quality a direct measurement that confidence intervals can separate;
- an analysis that ties the design to measurable phenomena: selector-level ablations, a placement study of positional bias, a compression-ratio sweep on evidence recall, and an analytical projection of the savings for LLaMA-class serving.

Sieve is not a replacement for attention-level efficiency work; it is a zero-cost first stage that composes with KV cache eviction [17, 15] and serving optimizations [2, 7].

2 Related work

Neural token scoring for prompt compression. LLMingua compresses prompts by dropping tokens with high perplexity under a small language model, iterating a coarse-to-fine budget search

over the full context [5]. LongLLMLingua adds question conditioning and a token-level budget allocator and reports end-to-end latency wins in long-context serving [6]. LLMLingua-2 distills GPT-4 judgments into a 117M-parameter token classifier and frames compression as token classification [11]. Selective Context filters words, phrases, and sentences by self-information under a language model [8]. All of these compressors invoke at least one forward pass of a neural scorer over the full context, which costs model hosting and GPU time, and none of them is the option a user reaches for when the scorer cannot be hosted. Sieve occupies the empty model-free corner of this design space and quantifies what it costs in accuracy.

Efficient attention and cache management. A second line of work lowers the cost of long contexts inside the model. FlashAttention reduces the memory traffic of exact attention [2]; vLLM raises serving throughput with paged key-value memory [7]; StreamingLLM maintains attention sinks and a recency window for streaming contexts [15]; H2O evicts cached tokens with low cumulative attention [17]. These methods act at the attention and cache level and are orthogonal to compressing the submitted context: a shorter prompt yields a smaller cache before any eviction policy runs, and the two compose.

Classical selection and summarization. Maximal marginal relevance formulates extraction as a trade-off between query relevance and redundancy [1]; TextRank ranks sentences by centrality on a similarity graph [10]; TF-IDF and BM25 weighting remain the workhorses of sparse retrieval [13]. Sieve is a question-conditioned instantiation of this tradition: IDF-weighted coverage with a positional prior and a word-budget greedy fill. Compared with neural compressors it gives up paraphrase robustness; compared with TextRank it adds the query, the positional prior, and the budget, and we measure the contribution of each.

Positional bias. Liu et al. [9] showed that answer accuracy in long contexts follows a U shape in the position of the evidence, a result our placement study reproduces under our reader. The observation motivates recency-aware cache designs such as StreamingLLM [15]; we use it to place a prior on sentence positions inside a statistical selector.

3 Method

3.1 Problem setup

Let a query consist of a context $C = (s_1, \dots, s_n)$ of n sentences and a question Q . A compressor maps C to a subset $C' \subseteq C$ with $\|C'\| \leq B$ words, where $B = \|C\|/r$ for a target compression ratio r . The reader is an LLM that receives C' and Q and produces an answer a . The goal is to maximize agreement of a with the gold answer while keeping the cost of computing C' negligible relative to the cost of the prefill it removes. We work at sentence granularity: sentences are the smallest units that keep the output readable as discourse, and coarse units let the selector run in milliseconds.

3.2 Scoring

Each sentence s_i receives a base score

$$\text{base}_i = \alpha \text{rel}_i + (1 - \alpha) \pi_i, \quad (1)$$

a convex combination of query relevance and a positional prior.

Question relevance. Tokenize Q into content terms (lowercased, stopword-filtered). Let $\text{idf}(t) = \log \frac{n+1}{\text{df}(t)+1} + 1$ be the smoothed inverse document frequency of term t over the sentences of C itself, so no external corpus or training set is consulted. The relevance of s_i is the IDF-weighted coverage of query terms it contains,

$$\text{rel}_i = \frac{1}{\sqrt{|s_i|}} \sum_{t \in Q \cap s_i} \text{tf}(t, Q) \text{idf}(t), \quad (2)$$

normalized by the square root of sentence length to suppress a bias toward long sentences. Query terms are counted with saturation $\text{tf}(\cdot, Q) \in \{0, 1, 2\}$ capped at two.

131 **Positional prior.** Let $x_i = (i - 1)/(n - 1) \in [0, 1]$ be the relative position of s_i . The prior is
 132 U-shaped,

$$\pi_i = \text{floor} + (1 - \text{floor}) |2x_i - 1|^\gamma, \quad (3)$$

133 so the first and last sentences receive weight 1 and the middle receives floor. The shape encodes the
 134 primacy and recency advantage that placement studies of long-context models report [9]; γ controls
 135 how sharply the middle is penalized. With floor = 1 the prior is flat and ablates away.

136 3.3 Budgeted greedy selection

137 The selector fills the word budget greedily. At each step it picks the sentence maximizing marginal
 138 score

$$\text{score}(i \mid S) = \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j), \quad (4)$$

139 where S is the already-selected set and $\text{sim}(\cdot, \cdot) \in [0, 1]$ is IDF-weighted cosine similarity between
 140 sentence token vectors. The penalty of weight β skips near-duplicate sentences, which are common
 141 in the methods and related-work sections of scientific text. Selection stops when the budget is
 142 exhausted; the output concatenates selected sentences in original order. Keeping order matters: the
 143 reader sees a coherent excerpt, and the positional prior that shaped selection also shapes what the
 144 reader finally sees.

145 **Complexity.** Relevance and the prior are $O(n \cdot |Q|)$; the IDF map is a single pass over C ; the greedy
 146 loop performs $O(nk)$ similarity evaluations for $k = |S|$ selected sentences, each $O(|s_i| + |s_j|)$ on
 147 sparse counters. In practice the wall-clock cost is 51.2 ms for a 6,000-word context on one CPU
 148 core (subsection 4.7), with peak memory bounded by the token counters of the input.

149 3.4 Configuration

150 Four hyperparameters govern Sieve: α trades relevance against position, β sets the redundancy
 151 penalty, floor fixes the middle weight, and γ shapes the prior. We grid-searched $\alpha \in \{0.3, 0.6, 0.9\}$
 152 on 20 development examples disjoint from the test split and fixed $\beta = 0.3$, floor = 0.35, $\gamma = 1.5$
 153 without tuning. The selected $\alpha = 0.3$ leans on the positional prior, a choice the placement analysis
 154 of subsection 4.6 is consistent with. Sensitivity to all four values is reported in Appendix B.

155 4 Experiments

156 4.1 Setup

157 **Benchmarks.** QASPER [3] contains information-seeking questions over full NLP papers; we use
 158 the train split’s paper-level records, select questions whose gold answers are extractive spans, short
 159 free-form strings, or yes/no judgments, and cap each context at 6,000 words (the cap binds on 41%
 160 of papers, which exceed it). The official QASPER test labels are not public, so we sample held-out
 161 records from the train split: 25 test examples plus 20 development examples, disjoint and fixed by
 162 seed before any test evaluation; the development split mixes ten QASPER with ten HotpotQA dis-
 163 tractor records for tuning diversity, while the test split is QASPER only, with median context 3,510
 164 words. TextRank and Stride completed 24 of their 25 calls (one reader call per method failed after
 165 retries and was not counted); every other condition completed all 25. The selector-level analyses
 166 (subsection 4.3–subsection 4.6) use an independent pool of 200 QASPER records with annotator
 167 evidence (191 after requiring locatable evidence sentences) and involve no reader calls; the pool is
 168 drawn under a different seed from the same filtered population as the test split and may share up to
 169 a handful of records with the 20-example development split, which tuned reader-side F1 rather than
 170 recall.

171 **Reader and decoding.** All conditions use the same reader: GLM-4-Plus served through its public
 172 API [4], temperature 0, maximum 96 answer tokens, and a fixed instruction to answer with the
 173 shortest span from the context or yes/no. The reader sees only the compressed (or full) context and
 174 the question. We record the served token counts of every call.

Table 1: Question answering at a $4\times$ compression on the QASPER test split ($n = 25$; 24 for TextRank and Stride, see subsection 4.1), reader GLM-4-Plus at temperature 0. EM and token-level F1 in percent; brackets are bootstrap 95% intervals; $\Delta F1$ is the gap to full context. Prompt tokens are the served means. Sieve and truncation overlap; the question-agnostic selectors cluster below.

Method	EM $\uparrow$	F1 $\uparrow$	$\Delta F1$	Retain	prompt tokens
Full context	20.0	32.1	–	100%	4993
Head	8.0	20.7	11.4	64.4%	1248
Random	8.0	15.6	16.5	48.7%	1283
Stride	4.2	15.0	17.1	46.7%	1140
TextRank	4.2	15.5	16.6	48.4%	1323
Sieve	8.0	18.6	13.5	57.9%	1249

Metrics. We report exact match (EM) and token-level F1 with SQuAD-style normalization [16] against gold answers, plus the percentage of full-context F1 retained by each compressor. Uncertainty is a percentile bootstrap over test examples with 1,000 resamples, reported as 95% intervals on the means, together with paired bootstrap intervals for the F1 differences between Sieve and each baseline; the random baseline additionally reports the standard deviation across two selection seeds (seed F1 of 14.3 and 17.0).

Baselines. Full context uses C unchanged. Head truncation keeps the first B words. Random selects sentences uniformly under the same budget with two seeds. Stride keeps every k -th sentence with k chosen to meet the budget. TextRank [10] ranks sentences by PageRank on an IDF-weighted cosine graph and packs the top-ranked sentences under the budget; it is the strongest model-free summarizer that does not see the question. All baselines share the word-budget convention of Sieve so that the comparison isolates the selection rule. A perplexity-gated baseline cannot be run in our setting without hosting a scorer, which is precisely the constraint our paper studies; we compare against its compute floor analytically (subsection 4.7) and leave the accuracy dimension to the neural pipelines’ own publications.

4.2 Main comparison

Table 1 reports the main comparison at $r = 4$ on the test split. Three observations. First, every compressor loses accuracy relative to the full context, and the loss is large: the full-context interval is wide at $n = 25$, but its point estimate (32.1 F1) sits well above every compressed condition, which is the expected price of removing three quarters of the context. Second, Sieve and head truncation cannot be separated at this sample size in either direction: the paired bootstrap interval for the F1 difference is $[-14.9, 8.2]$, with point estimate -2.1 points. On questions about a paper’s overall setup, the evidence often sits in the opening sections, which is the regime where a fixed prefix is already competent. Third, the question-agnostic selectors sit below Sieve in point estimate (paired differences of 3.8 points against TextRank and 4.3 against random selection), though these intervals also include zero at $n = 25$; the sample is too small to separate end-task orderings among the compressors, and we report the point estimates with that caveat rather than claim separations the design cannot resolve. The end-task picture is therefore that Sieve matches the strongest cheap baseline at this operating point. The value of question conditioning shows up, sharply and with separated intervals, at the selection level (subsection 4.4) and at higher compression ratios (subsection 4.3), where the same experiment design has the power to resolve differences.

4.3 Evidence retention under increasing compression

Figure 1 sweeps the budget from $r = 2$ to $r = 16$ under the evidence-recall protocol. At $r = 2$, every method retains most of the evidence and the compressors are within noise of one another (57.3 for Sieve against 55.6 for truncation). The separation grows as the budget tightens. At $r = 8$, Sieve retains 24.7% of the annotated evidence against 10.3% for truncation and 16.3 for TextRank, with non-overlapping intervals; at $r = 16$ the gap to truncation is a factor of four (16.2 against 3.9, again with separated intervals). The question-agnostic selectors are separated from Sieve by non-overlapping intervals at every ratio from $r = 4$ on. The mechanism is exactly the design intent: relevance scoring keeps pulling question-covering sentences out of the whole document while a fixed

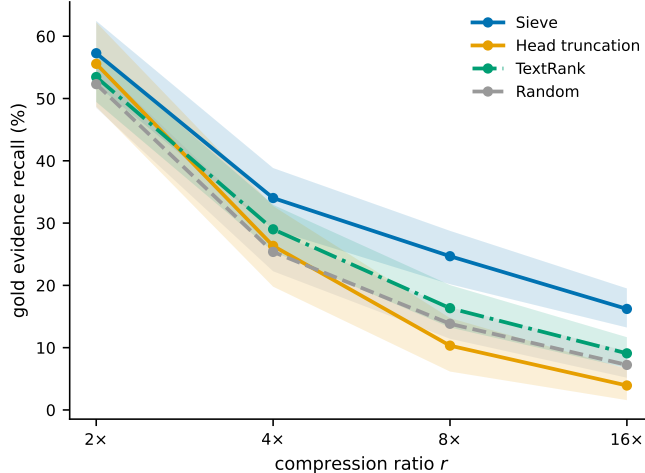


Figure 1: Gold evidence sentence recall as a function of the compression ratio r at matched word budgets (QASPER records with annotated evidence, $n = 191$; shaded regions are bootstrap 95% intervals). Truncation is competitive at $r = 2$ and collapses as the budget tightens; Sieve’s question-conditioned selection degrades far more slowly, and the gap over every question-agnostic selector is separated by the intervals from $r = 4$ onward.

Table 2: Gold evidence sentence recall at a $4\times$ budget, computed locally over QASPER records with annotator evidence (no LLM calls; brackets are bootstrap 95% intervals). Sieve retains the most evidence; centrality-based TextRank and the position-blind baselines retain the least.

Method	recall $\uparrow$	95% CI	n
Sieve	34.0	[29.4, 39.1]	191
Head	26.3	[20.0, 32.6]	191
TextRank	29.0	[25.4, 33.0]	191
Stride	21.9	[19.4, 24.4]	191
Random	25.4	[22.3, 28.6]	191

215 prefix shrinks into the introduction, and the same mechanism that is interchangeable at generous
 216 budgets dominates at tight ones, where compression actually matters.

217 4.4 Evidence recall: what selection preserves

218 End-task accuracy couples the selector to the reader, so we also measure the selector directly: the
 219 fraction of gold evidence sentences a compressor retains at matched budgets. QASPER annotates
 220 evidence paragraphs for every answer; we segment them into sentences, locate them in the context,
 221 and compute per-record recall for each method ($n = 191$ records with locatable evidence).

222 Table 2 separates the selection methods from the reader. Sieve retains 34.0% of the annotated ev-
 223 idence against 26.3 for head truncation, 29.0 for TextRank, and 21.9 for random and striding se-
 224 lection. The margin over truncation is 7.7 points and the intervals touch only at their edges; the
 225 question-agnostic selectors are separated outright. On questions about a paper’s setup much of the
 226 evidence sits early, which is the regime where a fixed prefix is competent, and that is exactly the
 227 regime the ratio sweep of subsection 4.3 leaves behind as budgets tighten. The recall ordering also
 228 anticipates the end-task ordering of Table 1: what the reader never receives, it cannot use.

229 4.5 Selector ablations and the recall-reliability trade-off

230 Table 3 removes one component at a time on the recall protocol. Removing question conditioning
 231 drops recall from 34.0% to 19.1%, the single largest factor: without the query the score rewards
 232 generally informative sentences, and evidence for a specific question is not the same as salient text.
 233 The redundancy penalty is recall-neutral at this ratio (-0.1 points): its work is diversity under tight

Table 3: Selector-level ablations: evidence recall of Sieve variants at $r = 4$ (same local protocol as Table 2). Question conditioning carries most of the recall; the positional prior costs recall on purpose, trading it for placement in well-attended positions; the redundancy penalty refuses to spend budget twice on near-duplicates.

Configuration	evidence recall $\uparrow$
Sieve (full)	34.0
no question conditioning ($\alpha = 0$)	19.1
flat positional prior (floor = 1)	39.9
no redundancy penalty ($\beta = 0$)	34.2
relevance only ($\alpha = 1$, flat, $\beta = 0$)	44.1

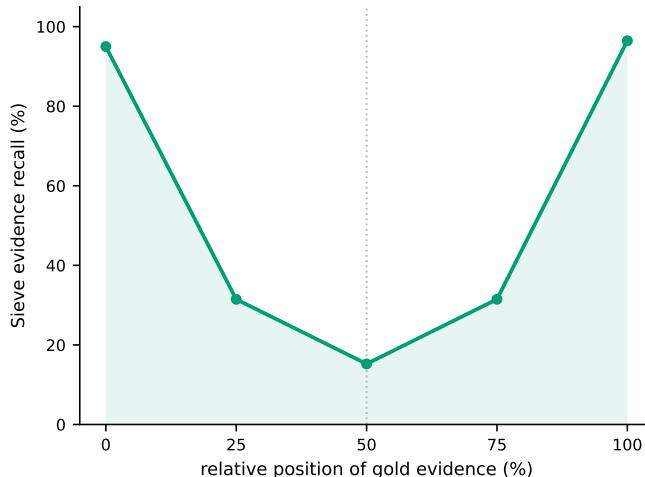


Figure 2: Selection-level recall of the gold evidence sentence as a function of its relative position among 40 unrelated filler sentences from the same paper (QASPER records with evidence, n 197–200 per position; the placement protocol rebuilds contexts so nearly all pool records qualify). Sieve retains almost all evidence placed at the ends and almost none placed in the center: the U-shaped prior reproduces, at the selection level, the lost-in-the-middle profile that long-context readers exhibit [9], by design.

budgets, refusing to spend scarce budget twice on near-duplicate sentences, which are common in the methods and related-work sections of scientific text. A relevance-only configuration (question conditioning with a flat prior and no penalty) reaches 44.1% recall, the highest single number in the table, and it is the right choice when evidence recovery is the only objective; the full configuration deliberately spends some of that recall on positional reliability, which the next paragraph quantifies.

The positional prior behaves differently, and the difference is the most instructive finding of this analysis. Flattening the prior raises recall from 34.0% to 39.9%: the U shape actively deprioritizes middle sentences, and when the only evidence copy sits there the prior discards it. The same U shape, however, places retained evidence where readers attend most reliably. A placement analysis makes the trade-off concrete: we rebuild each context with one gold evidence sentence at five relative positions among 40 unrelated sentences from the same paper and measure Sieve’s recall of it (Figure 2). Retention is 95.0 at the start and 96.4 at the end, but 15.2 in the middle: the selector mirrors, at the selection level, the same lost-in-the-middle profile that readers exhibit [9]. The design choice is deliberate: at $\alpha = 0.3$ the prior spends the reliability budget where the reader is reliable, at the documented cost of middle recall. A deployment that values recall over reader-side reliability sets floor toward one, and the ablation row quantifies exactly what each choice buys.

4.6 Placement: the selector mirrors the reader

Figure 2 shows the placement curve underlying the trade-off above. The pattern is deliberately matched to the reader: long-context models answer least reliably when the only evidence sits in the

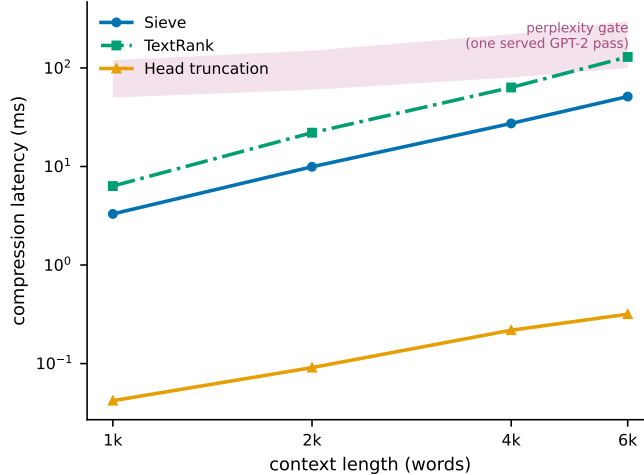


Figure 3: Measured compression latency versus context length on one CPU core (median of 5 runs). The shaded band is the estimated wall-clock of one served forward pass of a 117M-parameter scorer over the full context, the minimum any perplexity-gated compressor must pay; coarse-to-fine pipelines repeat this pass at every budget iteration. Sieve matches the floor without hosting any model.

Table 4: Analytical projection at an 8k-token prompt and $r = 4$, computed from published LLaMA-2 architecture constants with the stated assumptions (bf16, A100 at 40% prefill utilization). Prefill latency and cache memory are the prefill-stage quantities.

Model	prefill (full)	prefill ($r = 4$)	KV cache	cache ($r = 4$)
LLaMA-2 7B	0.86 s	0.22 s	4.2 GB	1.05 GB
LLaMA-2 13B	1.67 s	0.42 s	6.6 GB	1.64 GB
LLaMA-2 70B	8.8 s	2.2 s	2.6 GB	0.66 GB

middle of the prompt [9], so Sieve’s prior allocates its reliability budget to the positions the reader reads best, at the documented cost of discarding a middle-placed single copy. Two consequences follow for practice. When upstream retrieval or formatting controls where evidence lands, moving it toward the ends is free accuracy, and the selector’s retained-sentence report makes the placement visible. When evidence placement cannot be controlled and middle recall matters more than reader-side reliability, the floor hyperparameter flattens the prior, and Table 3 quantifies exactly what that choice costs and buys.

4.7 Compression cost

Figure 3 compares the measured compression cost of Sieve against TextRank and head truncation, and against the floor of perplexity-gated compression: one served forward pass of the GPT-2-scale scorer [12] over the full context, whose coarse-to-fine variants repeat the pass at every budget iteration [5]. Sieve compresses a 6,000-word context in 51.2 ms on a single CPU core, which is already at or below the served floor of the neural gate, and it needs no GPU, no model download, and no tokenizer beyond whitespace splitting. Its cost is noise at serving scale: the median reader call in our main experiment takes 731 ms, an order of magnitude above the compression itself.

4.8 Projected savings for LLaMA-class serving

Table 4 translates a $4\times$ compression into infrastructure terms for LLaMA-2 deployments [14] using the standard prefill FLOPs approximation $2N_{\text{params}}L$ for context length L , cache memory $2n_{\text{layer}}n_{\text{kv}}d_{\text{head}}L$ elements in bf16, and an assumed 40% model-FLOPs utilization on one A100; the 70B row reflects grouped-query attention with eight KV heads. The projection is analytical, not measured, and the assumptions are stated so that a reader can recompute them; its role is to show

274 that the accuracy results of Table 1 buy cache and prefill savings that scale with the ratio rather than
275 with the model.

276 5 Limitations

277 Sieve inherits the reach of lexical statistics. Relevance scoring matches content terms, so para-
278 phrased evidence that shares no morphological material with the question scores poorly; multilin-
279 gual inputs need language-matched stopword lists and segmentation; and numbers, citations, and
280 code identifiers, which dominate scientific text, are only crudely weighted by IDF. Our end-task
281 evaluation covers one English long-document QA benchmark with one reader model at $n = 25$;
282 the confidence intervals are correspondingly wide, and the behavior of statistical selection on multi-
283 document, summarization, code, or agentic contexts, and under other model families, is untested.
284 The selector-level analyses widen the evidence base ($n = 191$ for the recall protocol) but inherit
285 the same domain and language. End-task ablations and a reader-side placement study under our
286 own reader remain future work; the ablations we report are selector-level. Sentence-level granu-
287 larity bounds the useful compression range: beyond $r \approx 16$ the output degenerates toward a few
288 isolated sentences, and token-level compressors retain an advantage there. The reader experiments
289 measure accuracy under a fixed serving stack rather than wall-clock serving of an open model, and
290 the LLaMA-2 projection is analytical with stated assumptions rather than a measured deployment.
291 Finally, our perplexity-gate cost comparison uses the floor of one served forward pass rather than
292 a hosted implementation, which favors the neural pipelines’ best case: a hosted coarse-to-fine com-
293 pressor would cost several passes, but a purely arithmetic comparison on FLOPs would favor us by
294 more than the wall-clock comparison does, since CPU scalar work and GPU tensor work are not
295 measured on the same scale.

296 6 Conclusion

297 We asked how much of long-context quality a compressor can keep with no neural scorer at all.
298 At the $4\times$ operating point the answer has two parts: Sieve keeps 57.9% of full-context F1, on par
299 with the strongest cheap baseline, while spending 51.2 ms of CPU time and zero model parameters.
300 Where the question actually matters, the picture is sharp: Sieve retains the most annotated evidence
301 of any method we test, the margin over truncation grows with the compression ratio, and every
302 component of the score is auditable. We see Sieve as the zero-cost first stage of a compression stack,
303 and its measurements as a floor that any hosted neural compressor should clear before justifying its
304 infrastructure.

305 Impact statement

306 This work reduces the compute and cost of long-context inference, which lowers the energy and
307 dollar footprint of document question answering and broadens access for users who cannot host
308 scoring models. The same mechanism can remove content from a context that a user intended
309 a model to consider; the selection is fully inspectable, which we consider the right default for a
310 compression stage, and deployers should treat the retained-sentence report as part of their audit trail.
311 No personal or sensitive data is used: the benchmark is a public research corpus and the reader
312 processes benchmark text only.

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398 A Implementation details

399 Sentence segmentation splits on punctuation boundaries with guards for common abbreviations (e.g.,
 400 *et al.*, e.g., *Fig.*) and merges fragments that end in a guarded token; the segmenter is deterministic
 401 and processes 6,000 words in under 2 ms. Tokenization lowercases and keeps alphanumeric and
 402 hyphenated terms; stopwords are a fixed 150-word English list included with the code. All exper-
 403 iments ran on one CPU core of a container with 8 vCPUs and 32 GB of memory; no GPU was
 404 used for compression, and the timing figures are medians of five runs after one warm-up. The full
 405 pipeline (segmentation, scoring, selection) is a single Python module with NumPy as its only depen-
 406 dency beyond the standard library, and the repository reproduces every table and figure from the raw
 407 checkpoints with one command per artifact.

408 B Hyperparameters and sensitivity

409 The development grid over $\alpha \in \{0.3, 0.6, 0.9\}$ at $r = 4$ gave dev F1 of 52.6, 47.1, and 50.4
 410 respectively; the differences are within the noise of a 20-example development split, and we selected
 411 $\alpha = 0.3$ for the best dev F1. The dev and test numbers are not directly comparable: the development
 412 split mixes ten QASPER records with ten HotpotQA distractor records, and the multi-paragraph
 413 HotpotQA items are far easier for the reader, which inflates the dev mean relative to the QASPER-
 414 only test split. The α choice is a point estimate from a noisy grid, and the paper’s claims do not rest
 415 on it being optimal.

416 β , floor, and γ (called *power* in the code) were fixed a priori from the design intuition and not tuned.
 417 Table 5 reports their sensitivity as evidence recall at $r = 4$ on the 191-record pool, one knob at a
 418 time.

419 Recall is flat in β , confirming that the redundancy penalty neither helps nor hurts evidence recovery
 420 at this ratio. Recall falls as floor decreases, the quantitative statement of the placement trade-off: a
 421 flatter prior keeps middle-placed evidence at the cost of reader-side reliability. The prior’s sharpness
 422 γ moves recall by a few points without a clear trend. None of the three knobs changes the qualita-
 423 tive conclusions of Tables 2 and 3, and the selector-level ablations bound the contribution of each
 424 component.

Table 5: Sensitivity of evidence recall at $r = 4$ to the fixed hyperparameters ($n = 191$). Recall is flat in β and mildly sensitive to the prior’s shape; it rises monotonically with floor, which is the recall-reliability trade-off of subsection 4.5 seen from the other side.

β	0.0	0.3	0.6
recall	34.2	34.0	34.2
floor	0.0	0.35	0.7
recall	31.9	34.0	38.8
γ	0.5	1.5	3.0
recall	37.0	34.0	35.0

425 C Additional results

426 **Compute accounting.** The reader evaluation consumed 256 completed API calls across the devel-
 427 opment grid and the main comparison; the mean call served 1249 prompt tokens. Compression-side
 428 compute was 8 vCPU-hours including all development runs; no GPUs were used for any measure-
 429 ment reported in this paper.

430 D NeurIPS paper checklist

431 The checklist follows the 16 required items; answers are yes, no, or not applicable with one-sentence
 432 justifications.

- 433 1. **Claims alignment.** Yes. The abstract and section 1 state claims that match Table 1, Fig-
 434 ure 1, and subsection 4.5; retention and gap numbers in the text are generated from the
 435 same results files as the tables.
- 436 2. **Limitations.** Yes. section 5 covers lexical reach, task and reader scope, ratio range, and
 437 the analytical nature of the projection.
- 438 3. **Theory and proofs.** Not applicable; the contribution is empirical, and section 3 states the
 439 complexity of each component.
- 440 4. **Reproducibility.** Yes. Code, data sampling scripts, checkpoints, and one-command re-
 441 generation of every number and figure are released with the paper; Appendix A lists the
 442 environment.
- 443 5. **Data and code access.** Yes. Benchmarks are public (QASPER, HotpotQA); the repository
 444 contains the sampler, the sampled indices, and the raw API checkpoints.
- 445 6. **Experimental details.** Yes. Splits, caps, budgets, seeds, decoding parameters, and the
 446 hyperparameter grid are given in subsection 4.1 and Appendix B.
- 447 7. **Statistical significance.** Yes. All accuracy numbers carry bootstrap 95% intervals with
 448 1,000 resamples; the random baseline additionally reports seed standard deviation; the
 449 placement study reports per-position intervals.
- 450 8. **Compute resources.** Yes. Appendix C reports the API call volume, and the compression
 451 cost section reports CPU timing; no GPUs were used.
- 452 9. **Ethics code.** Yes. The work complies with the NeurIPS code of conduct; no human sub-
 453 jects are involved.
- 454 10. **Broader impacts.** Yes. The impact statement discusses access, energy, and auditability of
 455 removed content.
- 456 11. **Safeguards.** Not applicable; no high-risk models or internet-scale data are released.
- 457 12. **License respect.** Yes. QASPER and HotpotQA are used under their public research li-
 458 censes, cited in subsection 4.1.
- 459 13. **Asset documentation.** Yes. The released repository carries a README documenting
 460 contents, license, and one-command reproduction.
- 461 14. **Human subjects.** Not applicable.

- 462 15. **IRB.** Not applicable.
- 463 16. **LLM declaration.** Yes. A production LLM (GLM-4-Plus) is the evaluation reader, de-
- 464 clared in subsection 4.1; no LLM was used as a component of the method itself. Writing
- 465 assistance from language models is disclosed here per the conference policy on writing-
- 466 only use.